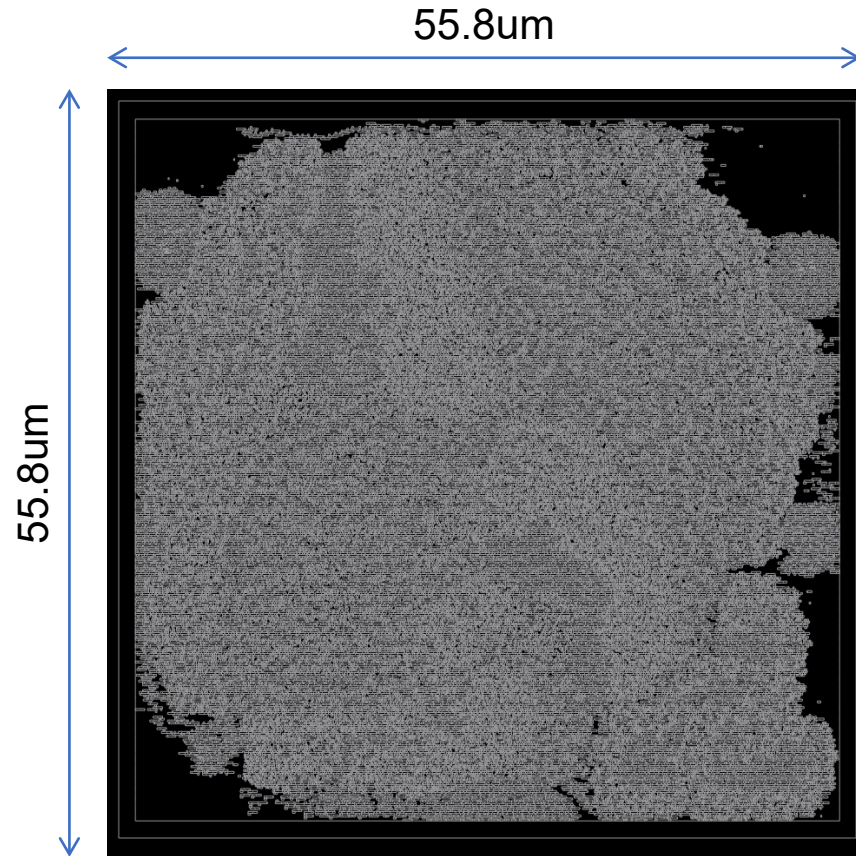
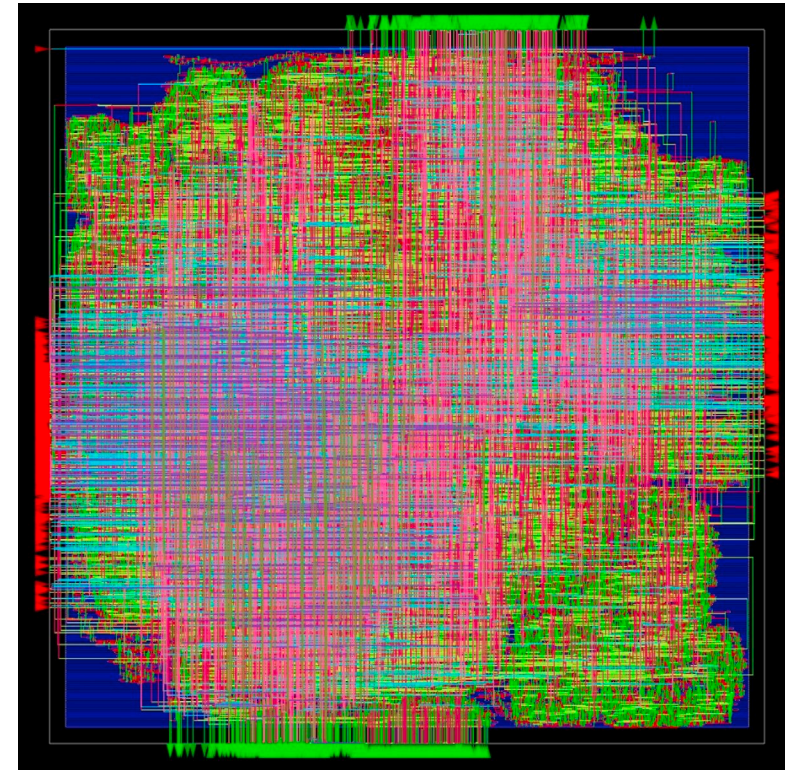


# Benchmark: ECG (62K Gates)

PDK: USC-3N-2D, EDA: OpenROAD



Placement



Routing

# Benchmark: ECG (62K Gates)

PDK: USC-3N-2D, EDA: OpenROAD

- Intended for PDK+EDA flow validation, not PPA optimization.

|             |                      |             |
|-------------|----------------------|-------------|
| Performance | Target freq (GHz)    | 2.00        |
|             | Effective freq (GHz) | 1.99        |
|             | WNS (ps)             | -1.30       |
|             | TNS (ps)             | -23.97      |
| Power       | Total power (mW)     | 22.01       |
|             | Internal power (mW)  | 16.64       |
|             | Switching power (mW) | 5.35        |
|             | Leakage power (mW)   | 0.03        |
| Area        | Chip dimension (um)  | 55.8 x 55.8 |
|             | Macro Count          | 0           |
|             | Cell Count           | 61,585      |
|             | Placement Density    | 72%         |

|                 |      |
|-----------------|------|
| Placement (min) | 0.63 |
| CTS (min)       | 0.12 |
| Routing (min)   | 13.9 |
| Total (min)     | 14.7 |

Turnaround time

|           |                    |
|-----------|--------------------|
| OS        | Red Hat Linux 8.10 |
| CPU       | Xeon Gold 6454S    |
| #Cores    | 64                 |
| CPU Speed | 3400 MHz           |
| Memory    | 512 GB             |

Server spec





# Benchmark: OpenPiton (292K Gates)

PDK: USC-3N-2D, EDA: OpenROAD

- Intended for PDK+EDA flow validation, not PPA optimization.

|             |                      |               |
|-------------|----------------------|---------------|
| Performance | Target freq (GHz)    | 1.00          |
|             | Effective freq (GHz) | 0.95          |
|             | WNS (ps)             | -54.34        |
|             | TNS (ps)             | -258.07       |
| Power       | Total power (mW)     | 101.87        |
|             | Internal power (mW)  | 94.75         |
|             | Switching power (mW) | 6.39          |
|             | Leakage power (mW)   | 0.72          |
| Area        | Chip dimension (um)  | 330.0 x 398.2 |
|             | Macro Count          | 28            |
|             | Cell Count           | 292,047       |
|             | Placement Density    | 72%           |

|                 |      |
|-----------------|------|
| Placement (min) | 9.5  |
| CTS (min)       | 2.1  |
| Routing (min)   | 32.4 |
| Total (min)     | 43.9 |

Turnaround time

|           |                    |
|-----------|--------------------|
| OS        | Red Hat Linux 8.10 |
| CPU       | Xeon Gold 6454S    |
| #Cores    | 64                 |
| CPU Speed | 3400 MHz           |
| Memory    | 512 GB             |

Server spec